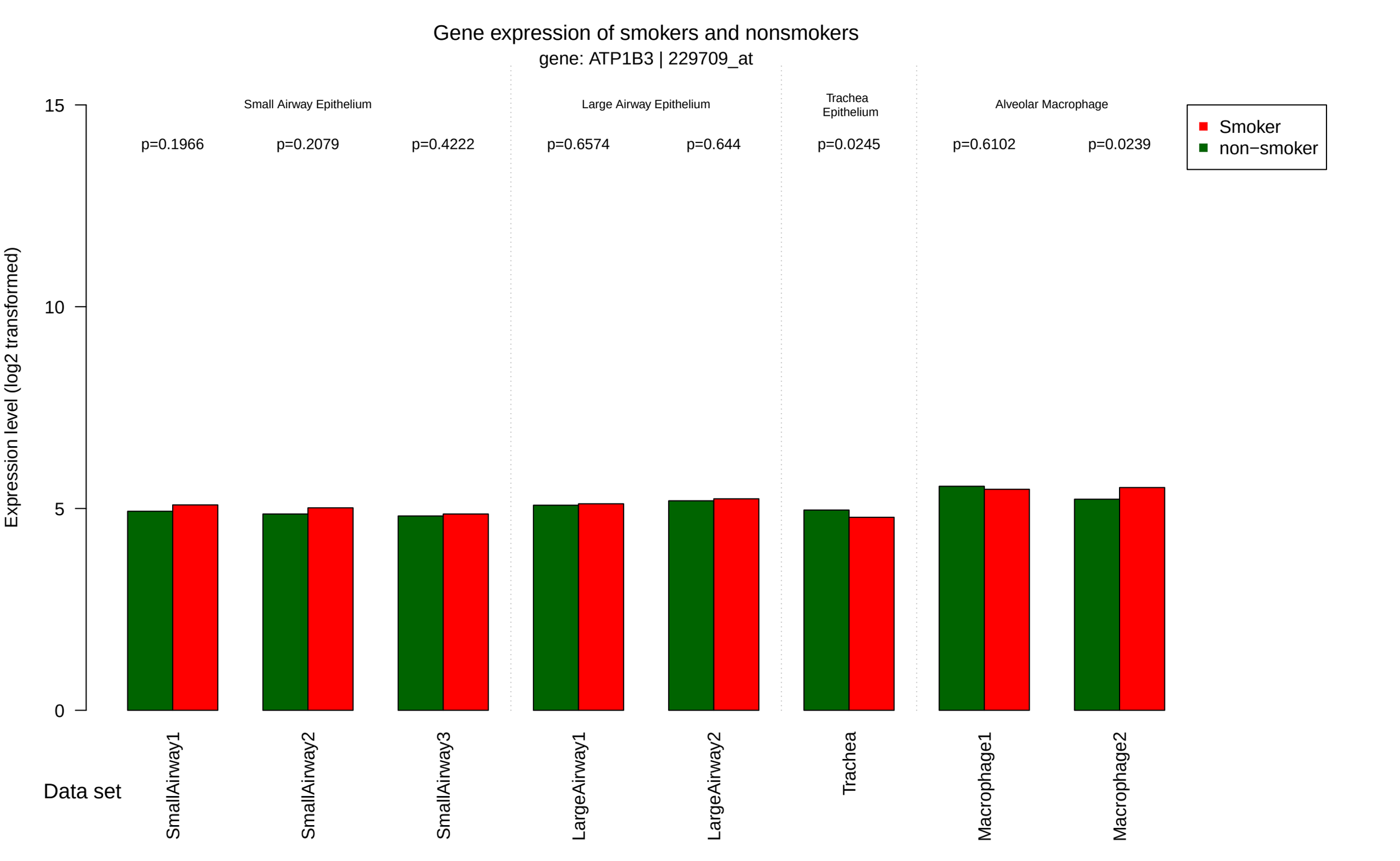
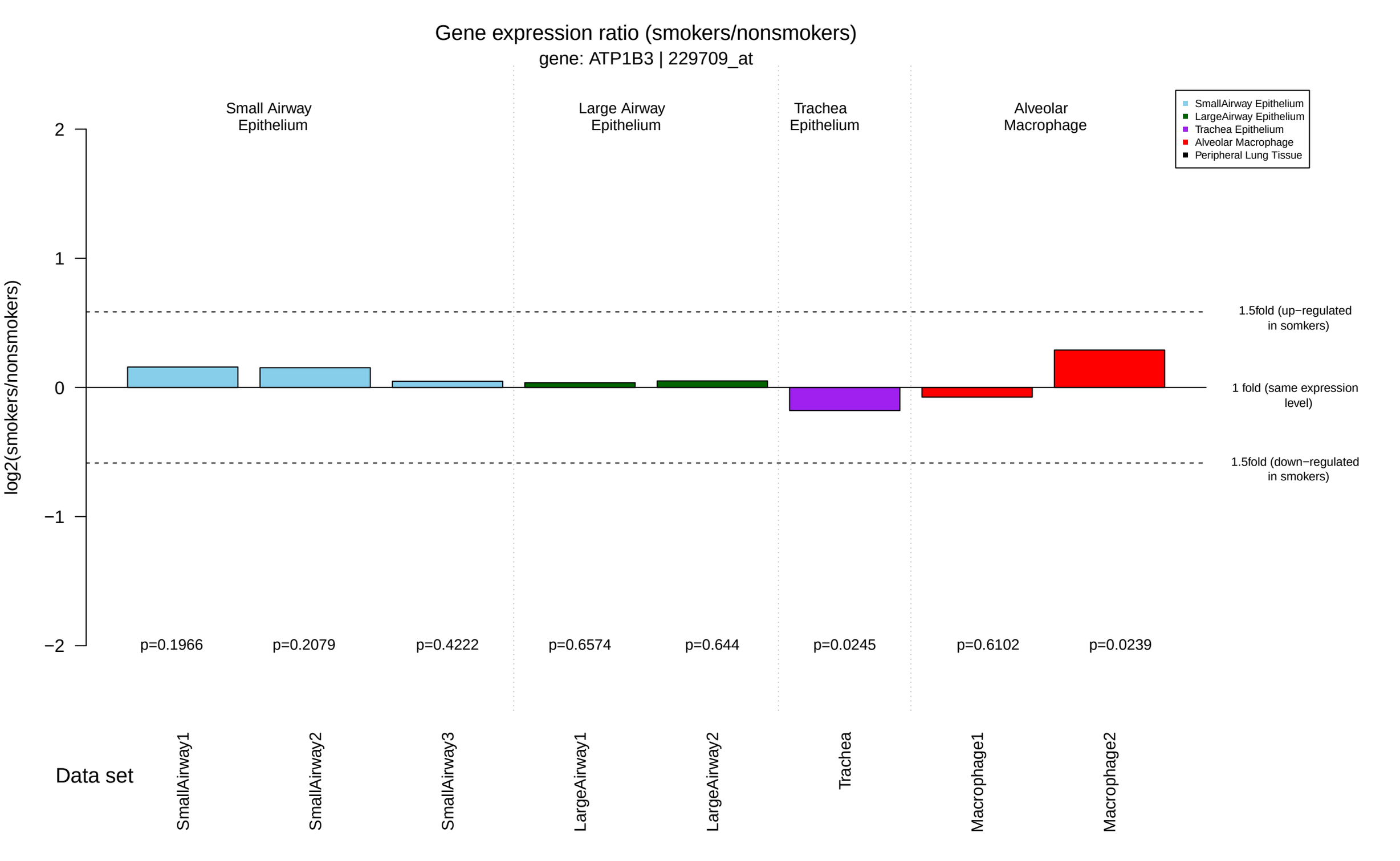
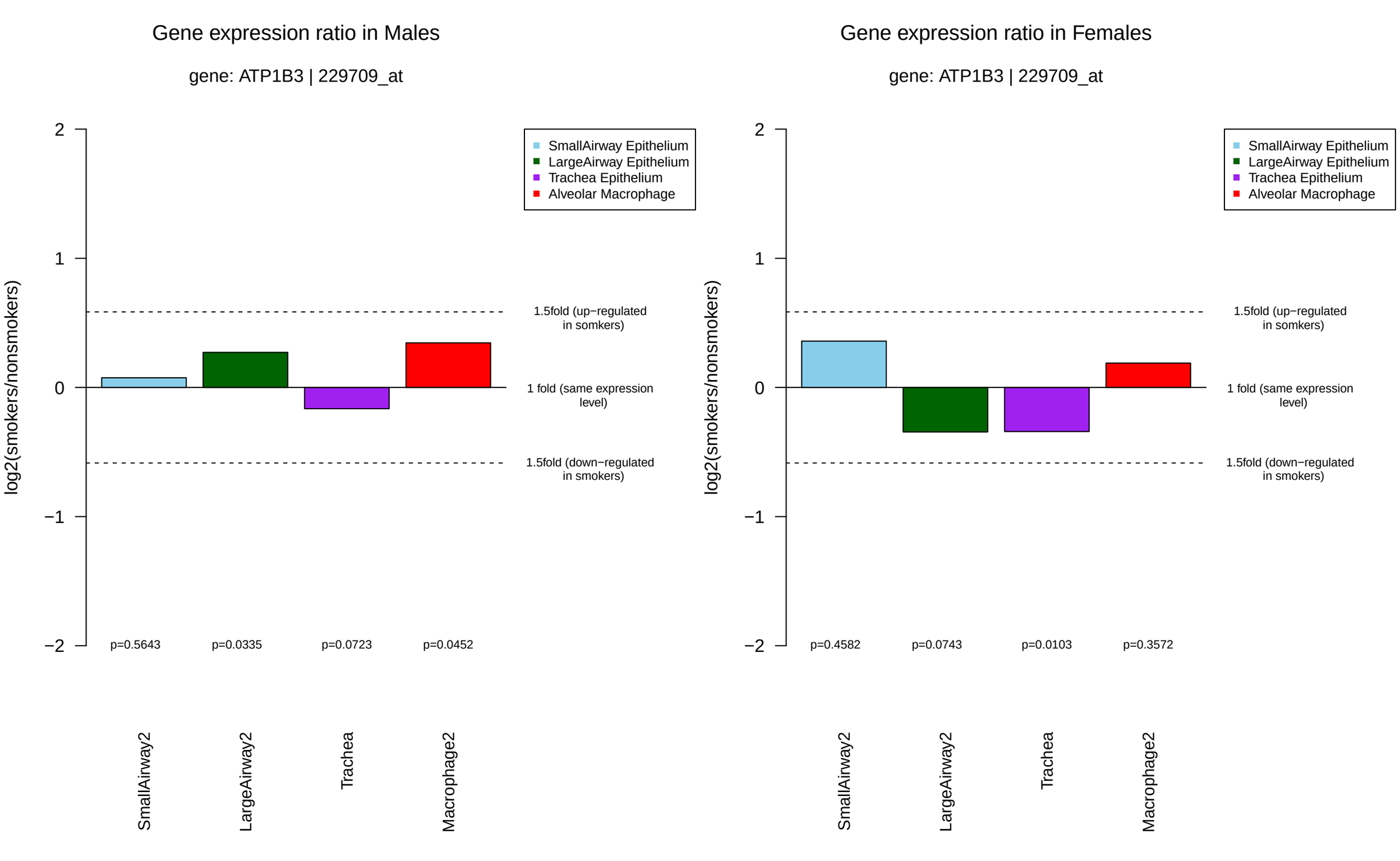
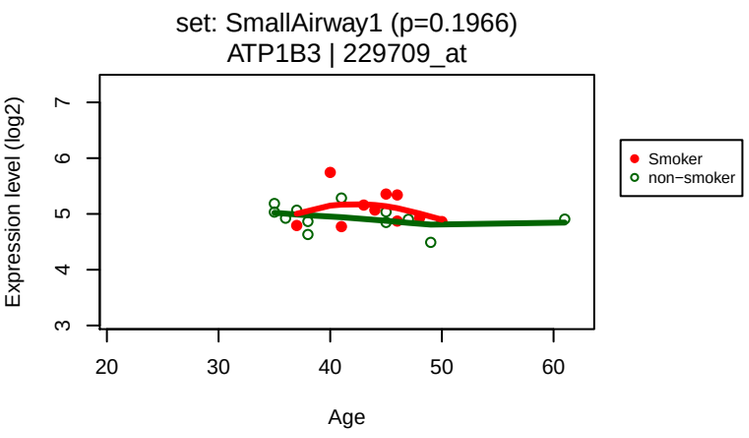
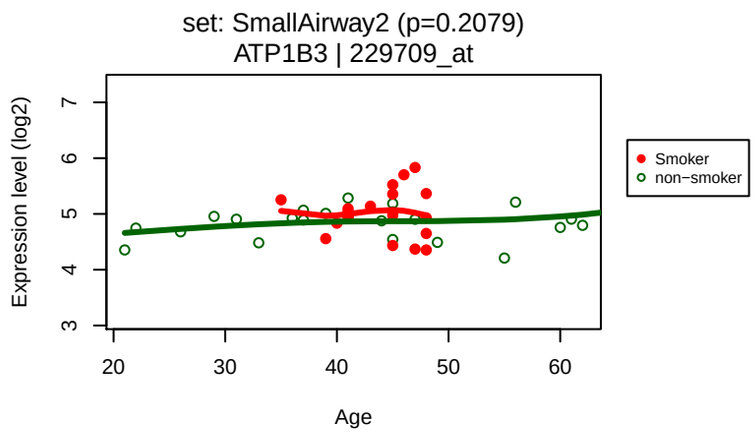
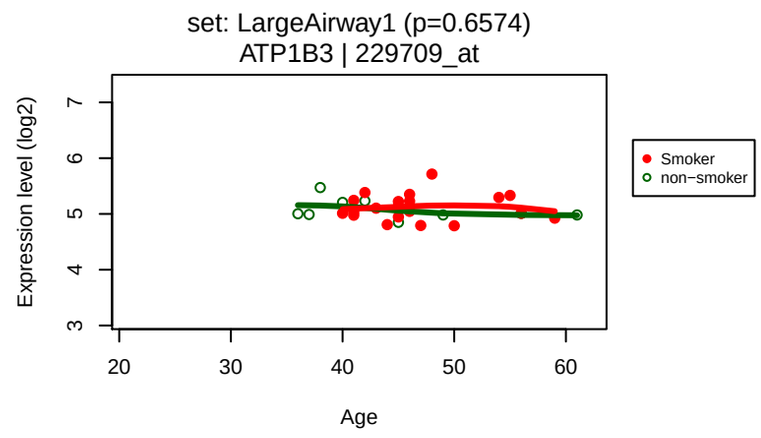

Gene expression of smokers and nonsmokers
gene: ATP1B3 | 229709_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1966
p=0.2079
p=0.4222
p=0.6574
p=0.644
p=0.0245
p=0.6102
p=0.0239
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

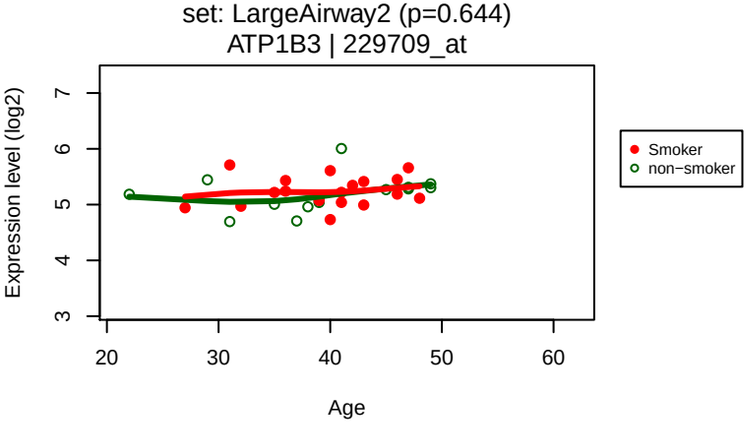

Gene expression ratio (smokers/nonsmokers)
gene: ATP1B3 | 229709_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1966
p=0.2079
p=0.4222
p=0.6574
p=0.644
p=0.0245
p=0.6102
p=0.0239
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

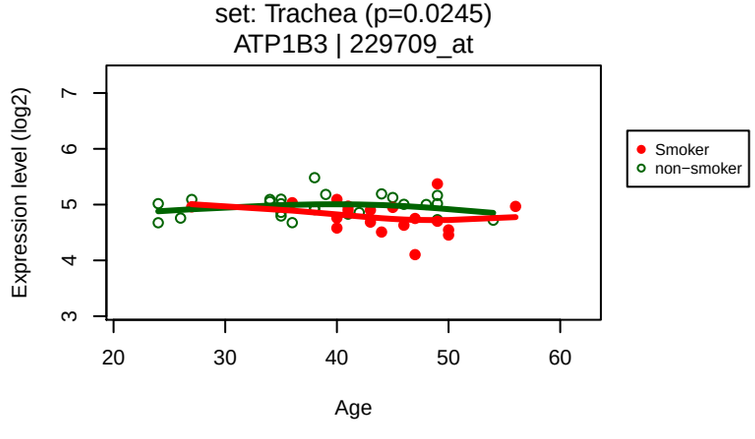

Gene expression ratio in Males
gene: ATP1B3 | 229709_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5643
p=0.0335
p=0.0723
p=0.0452
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP1B3 | 229709_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4582
p=0.0743
p=0.0103
p=0.3572
SmallAirway2
LargeAirway2
Trachea
Macrophage2

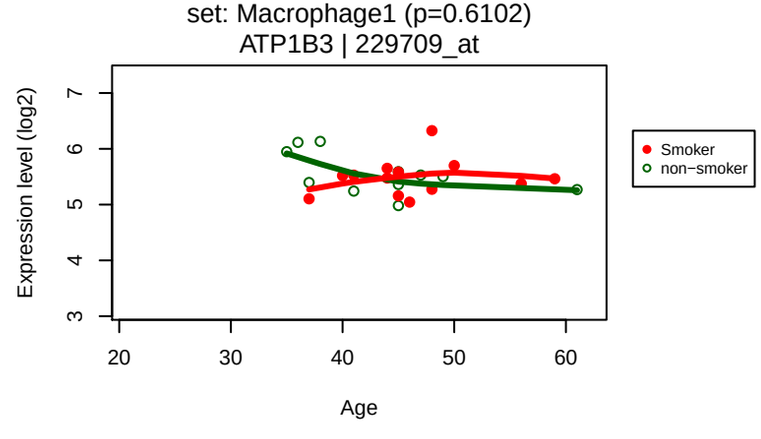

set: SmallAirway1 (p=0.1966)
ATP1B3 | 229709_at
Expression level (log2)
Age
Smoker
non−smoker

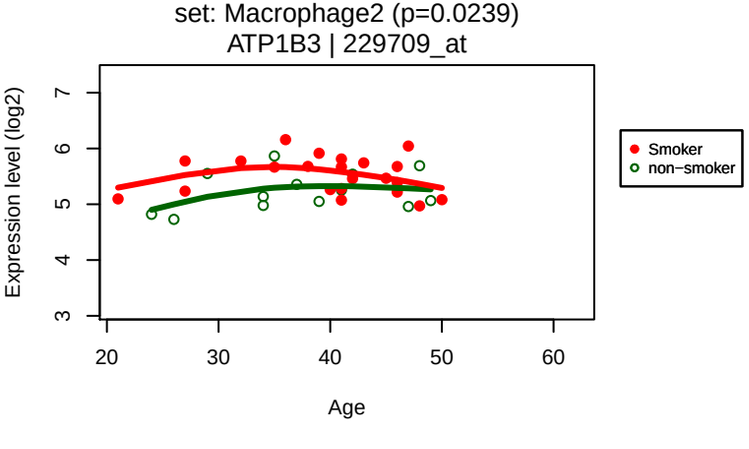

set: SmallAirway2 (p=0.2079)
ATP1B3 | 229709_at
Expression level (log2)
Age
Smoker
non−smoker

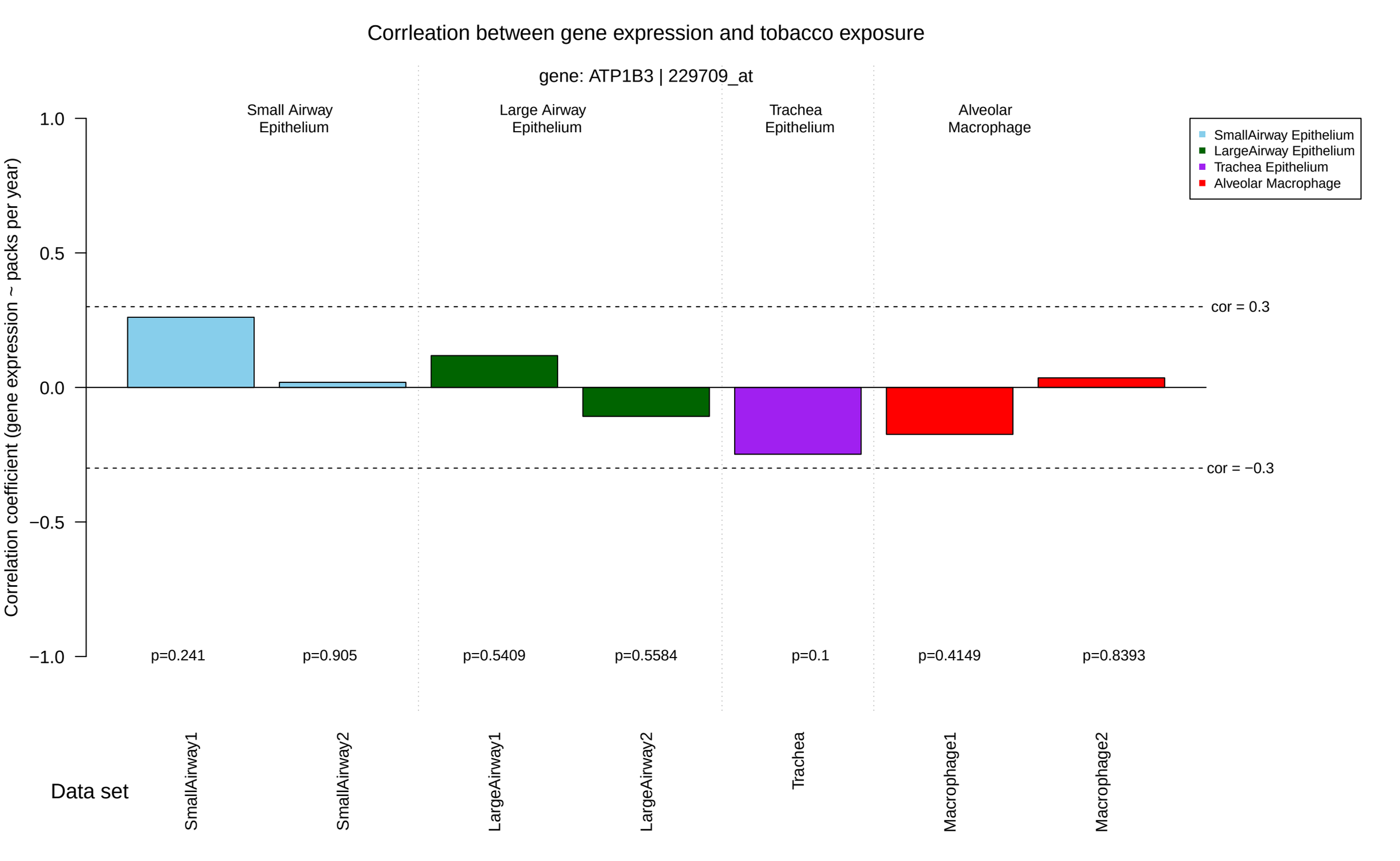

set: LargeAirway1 (p=0.6574)
ATP1B3 | 229709_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.644)
ATP1B3 | 229709_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0245)
ATP1B3 | 229709_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6102)
ATP1B3 | 229709_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0239)
ATP1B3 | 229709_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP1B3 | 229709_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.241
p=0.905
p=0.5409
p=0.5584
p=0.1
p=0.4149
p=0.8393
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2